

MODULE 4.3

# **Boolean Expressions and Logic Gate Applications**

## MODULE 4.3

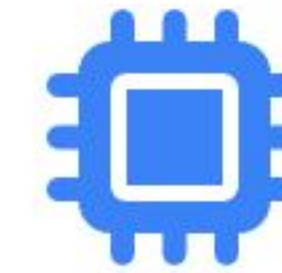
# Learning Objectives



**Write Expressions**



**Combine Gates**



**Create Circuits**



**Construct Truth Tables**



**Practical Applications**

# Boolean Expressions

Mathematical representations of logical operations.

**AND**

$$Q = A \cdot B$$

**OR**

$$Q = A + B$$

**NOT**

$$Q = \bar{A}$$

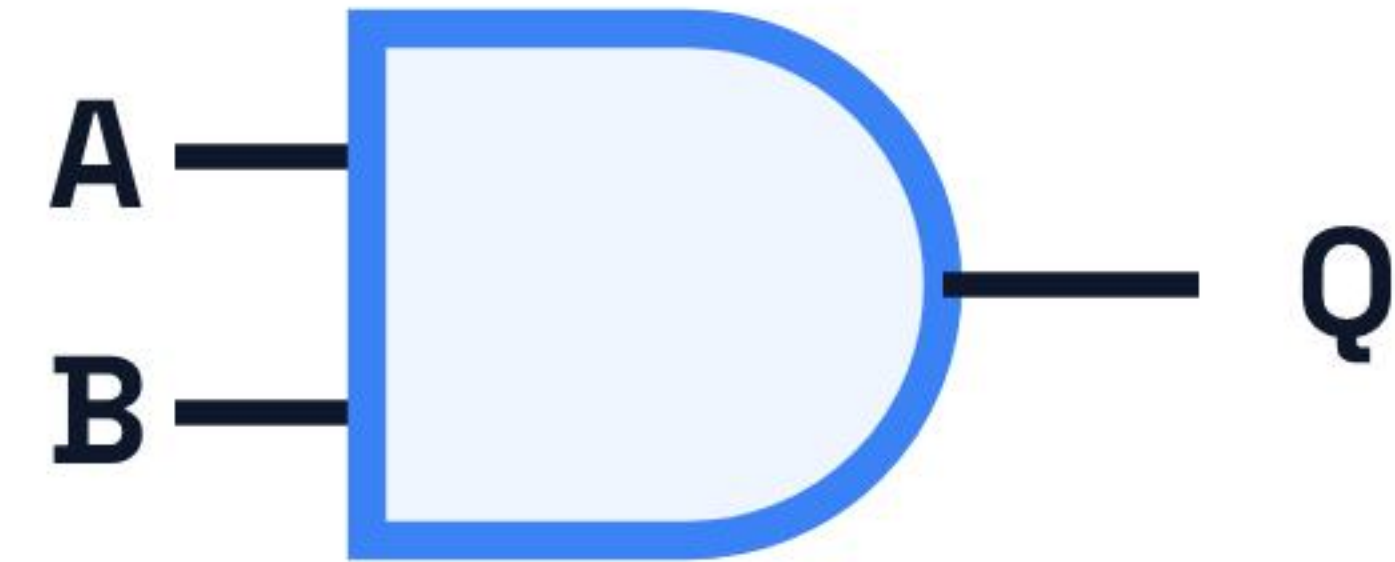
# From Expression to Circuit

EXPRESSION

$$Q = A \cdot B$$



LOGIC GATE



# Combining Logic Gates

Complex circuits use multiple basic gates working together.

$$Q = A \cdot (A + B) + B$$

**(A + B)**

OR Operation

**A · Result**

AND Operation

**Result + B**

OR Operation

# Reading Boolean Expressions

EXPRESSION

$$Q = A \cdot B + C$$

HOW TO READ IT

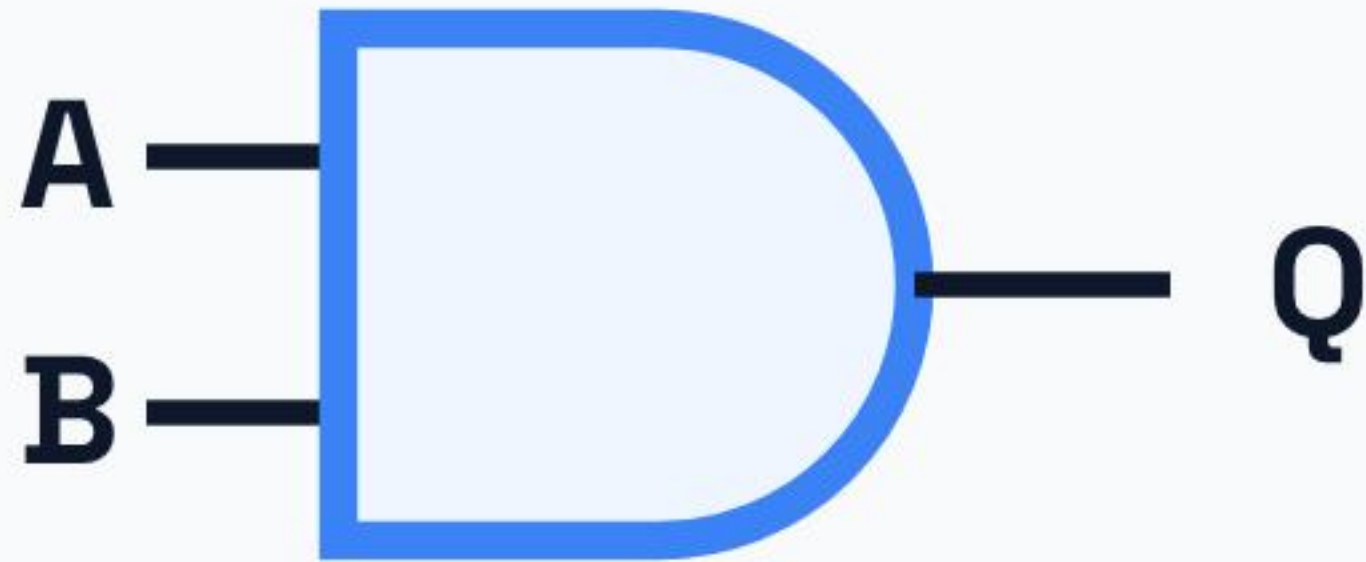
**A AND B OR C**

Step 1:  $(A \cdot B)$

Step 2:  $(A \cdot B) + C$

# From Circuit to Expression

LOGIC CIRCUIT



EXTRACTED EXPRESSION

$$Q = A \cdot B$$

# Combinational Logic

Creating complex logic circuits by linking standard gates.

**AND + NOT**



**OR + NOT**



**AND + OR**



# The NOR Gate

## NOT OR

Opposite of OR gate logic.

BOOLEAN EXPRESSION

$$Q = \overline{A + B}$$

# NOR Gate Truth Table

| A | B | $Q = \overline{A + B}$ |
|---|---|------------------------|
| 0 | 0 | 1                      |
| 0 | 1 | 0                      |
| 1 | 0 | 0                      |
| 1 | 1 | 0                      |

Output is 1 **only when ALL** inputs are 0.

# The NAND Gate

## NOT AND

Opposite of AND gate logic.

BOOLEAN EXPRESSION

$$Q = \overline{A \cdot B}$$

# NAND Gate Truth Table

| A | B | $Q = \overline{A \cdot B}$ |
|---|---|----------------------------|
| 0 | 0 | 1                          |
| 0 | 1 | 1                          |
| 1 | 0 | 1                          |
| 1 | 1 | 0                          |

Output is 0 **only when ALL** inputs are 1.

# Basic Gate Summary

| Gate | Meaning  | Output = 1 when...     |
|------|----------|------------------------|
| AND  | All      | All inputs = 1         |
| OR   | Any      | At least one input = 1 |
| NOT  | Opposite | Input = 0              |
| NOR  | NOT OR   | All inputs = 0         |
| NAND | NOT AND  | Not all inputs = 1     |

# Why Use Logic Gates?

Logic gates are the foundation of modern digital devices.



**Computers**



**Calculators**



**Mobile Phones**



**Televisions**

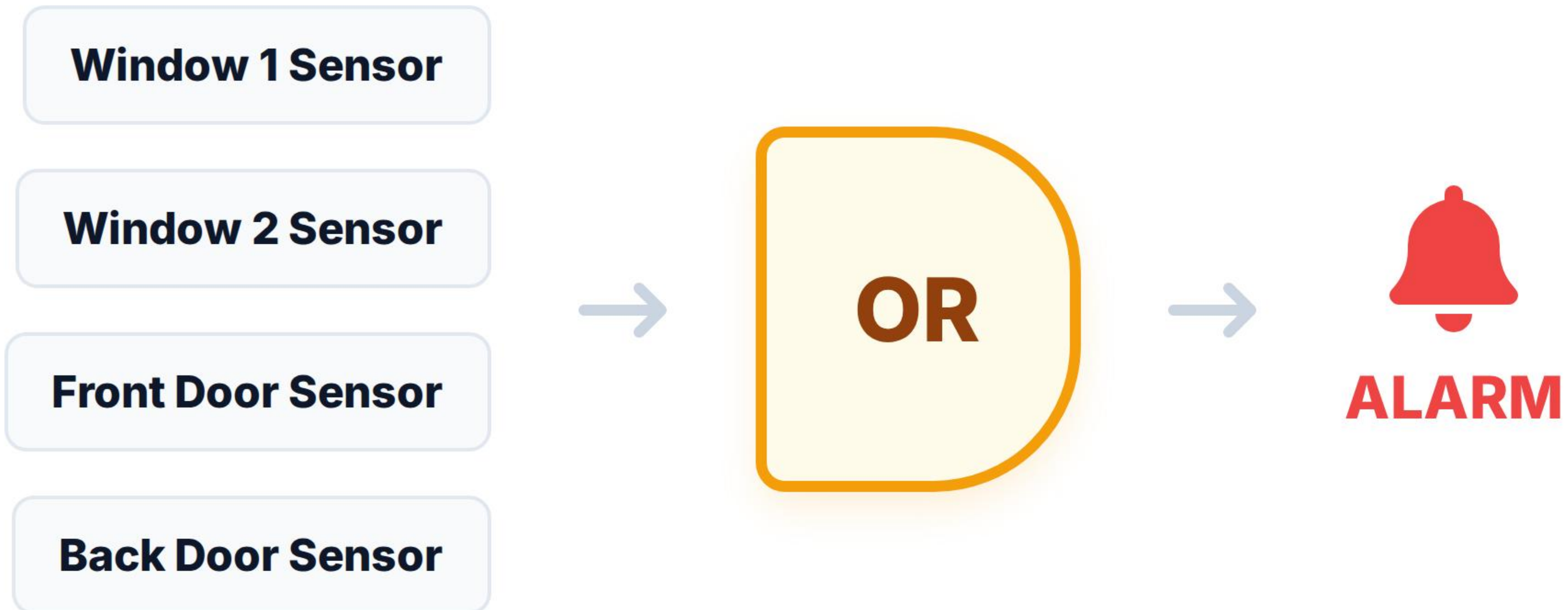


**Air Conditioners**



**Digital Clocks**

# Application: Home Alarm System



# Alarm System Logic

Any single sensor breach triggers the alarm.

Window 1 (Breached)

1

Window 2 (Secure)

0

Front Door (Secure)

0

Back Door (Secure)

0

**Result**

$$1 + 0 + 0 + 0 = 1$$

**ALARM ON**

# Application: Street Light Control

Logic circuits manage automated systems based on multiple conditions.



**Manual Switch**



**Time of Day Timer**



**Ambient Light Sensor**



## **Automated Decision**

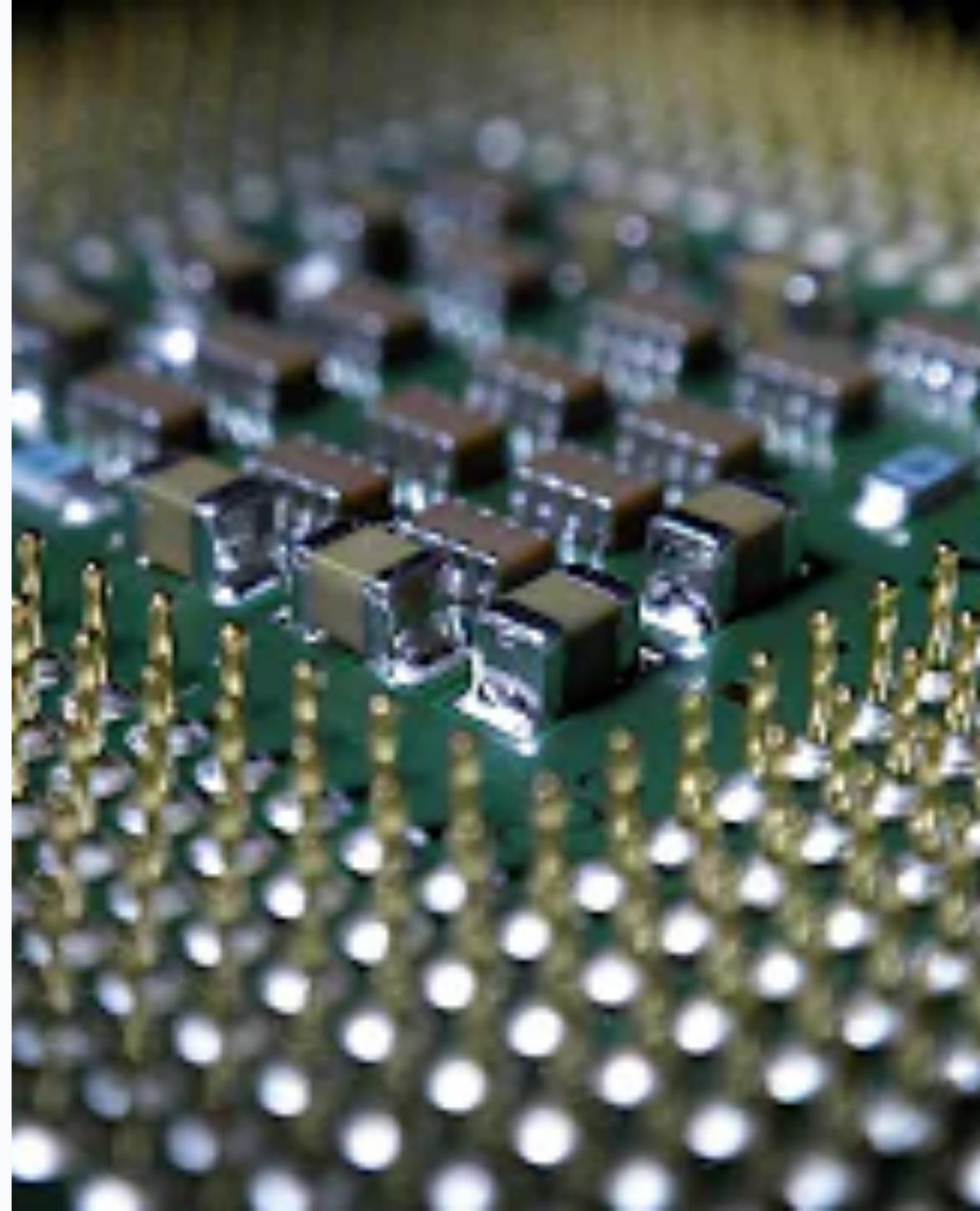
System determines ON/OFF state dynamically.

# Integrated Circuits (IC)

Small electronic circuits containing many components packed onto a single chip.

- ✓ **Transistors**
- ✓ **Resistors & Capacitors**
- ✓ **Diodes**

 **Logic Gates**



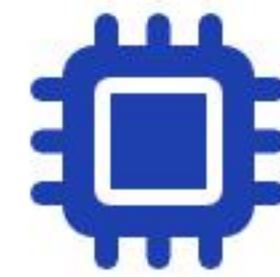
# Logic Gates in Computers

The fundamental building blocks of modern processing power.



## Logic Gates

Basic Decision Level



## Digital Circuits

Combinational Logic



## Computer System

Complete CPU Operations

# Remember These!

**AND**

**ALL** → 1

$$Q = A \cdot B$$

**OR**

**ANY** → 1

$$Q = A + B$$

**NOT**

**OPPOSITE**

$$Q = \bar{A}$$

**NAND**

**NOT AND**

**NOR**

**NOT OR**

# Final Truth Table Matrix

| A | B | AND | OR | NAND | NOR |
|---|---|-----|----|------|-----|
| 0 | 0 | 0   | 0  | 1    | 1   |
| 0 | 1 | 0   | 1  | 1    | 0   |
| 1 | 0 | 0   | 1  | 1    | 0   |
| 1 | 1 | 1   | 1  | 0    | 0   |

# Think & Answer

**1. What are the three basic logic gates?**

**2. What is the Boolean expression for AND?**

**3. When does an OR gate produce 1?**

**4. What does a NOT gate do?**

**5. How many combinations are there for 3 inputs?**

**6. What is a NAND gate made from?**

**7. What is a NOR gate made from?**

**8. Give two practical applications of logic gates.**

# Lesson Summary

- ✓ Explain digital signals and binary values.
- ✓ Define a logic gate.
- ✓ Identify AND, OR and NOT gates.
- ✓ Explain how each basic gate works.
- ✓ Construct basic truth tables.
- ✓ Write simple Boolean expressions.
- ✓ Identify NAND and NOR gates.
- ✓ Understand simple combinational circuits.
- ✓ Identify practical uses of logic gates.

LOGIC GATES

# The Decision Makers of Digital Systems

0 & 1 → Logic → Output

AND | OR | NOT | NAND | NOR



## Image Sources



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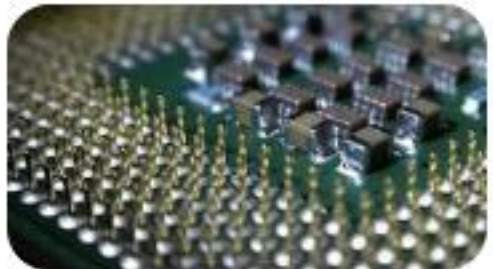
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